

# LaCie Drive Backup Plan

rsync Incremental Snapshots with Hardlink Deduplication

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|                       |                                                            |
|-----------------------|------------------------------------------------------------|
| <b>Server:</b>        | bunya (bunya.rcc.uq.edu.au)                                |
| <b>Remote path:</b>   | /QRISdata/Q1041/backups/                                   |
| <b>Source drives:</b> | /Volumes/LaCie_Top, /Volumes/LaCie_Bottom                  |
| <b>Log directory:</b> | ~/Documents/repos/scripts/backup_data_scripts/backup_logs/ |
| <b>Date:</b>          | March 2026                                                 |

# 1. How It Works

This backup strategy uses **rsync** with **hardlink-based incremental snapshots**. Each backup run creates a new dated directory on the server that looks like a complete copy of the drive. However, files that haven't changed since the last backup are stored as **hardlinks** to the previous snapshot, consuming essentially zero additional disk space.

Only new or modified files are transferred over the network. Deleted or renamed files are automatically handled. No manual tracking of folders, no Spotlight queries, no file lists.

Key advantages over the previous mdfind+tar approach:

- Automatically picks up new root folders, renames, and deletions
- Each snapshot is a complete browseable directory tree (no replaying deltas)
- Minimal network transfer and disk usage via hardlinks
- Simple restore: just rsync in the other direction

## 2. Remote Directory Structure

```
bunya:/QRISdata/Q1041/backups/
|-- snapshots_LaCie_Top/
|   |-- 2026-03-26/           <-- first full snapshot
|   |-- 2026-04-15/           <-- incremental (hardlinked)
|   |-- 2026-05-20/           <-- incremental (hardlinked)
|   +-- latest -> 2026-05-20  <-- symlink to most recent
+-- snapshots_LaCie_Bottom/
    |-- 2026-03-26/
    |-- 2026-04-15/
    +-- latest -> 2026-04-15
```

## 3. Initial Backup (Run Once)

The first backup transfers everything. Use `backup_initial.sh`. This creates the first dated snapshot and sets up the `latest` symlink.

```
chmod +x backup_initial.sh
caffeinate -s ./backup_initial.sh
```

What the script does for each drive:

1. Creates the remote snapshot directory via SSH
2. Runs rsync with `--partial --progress` (resumable if SSH drops)
3. Updates the `latest` symlink to point to the new snapshot
4. Saves a log file to the `backup_logs` directory

### Resume behavior:

Both scripts are **resume-safe**. If SSH drops mid-transfer, just re-run the same script. The script detects the incomplete snapshot folder on the server and resumes into it, rather than creating a new folder with today's date. The key mechanism:

- The latest symlink is only created **after** rsync completes successfully. An incomplete snapshot has a dated folder but no symlink update. On re-run, the script finds this orphaned folder and resumes into it.
- `rsync --partial` keeps partially transferred files on the server, so even a half-copied large file resumes from where it left off.
- If one drive finishes but the other doesn't, re-running skips the completed drive (it sees the latest symlink already exists).

### Core rsync command (for reference):

```
rsync -avz --partial --progress \
  --exclude='.DS_Store' --exclude='.fseventsd' \
  --exclude='.TemporaryItems' --exclude='.DocumentRevisions*' \
  --exclude='.Trashes' --exclude='.Spotlight*' \
  /Volumes/LaCie_Top/ \
  bunya:/QRISdata/Q1041/backups/snapshots_LaCie_Top/2026-03-26/
```

## 4. Incremental Backup (Every Future Run)

Use `backup_delta.sh` for all subsequent backups. The `--link-dest` flag tells rsync to hardlink unchanged files from the previous snapshot, and `--delete` ensures files removed from the drive are also removed in the new snapshot.

```
caffeinate -s ./backup_delta.sh
```

### Core rsync command (for reference):

```
rsync -avz --partial --progress --delete \
  --link-dest="/QRISdata/Q1041/backups/snapshots_LaCie_Top/latest" \
  --exclude='.DS_Store' --exclude='.fseventsd' \
  --exclude='.TemporaryItems' --exclude='.DocumentRevisions*' \
  --exclude='.Trashes' --exclude='.Spotlight*' \
  /Volumes/LaCie_Top/ \
  bunya:/QRISdata/Q1041/backups/snapshots_LaCie_Top/2026-04-15/
```

## 5. rsync Flags Explained

| Flag                     | Purpose                                                              |
|--------------------------|----------------------------------------------------------------------|
| <code>-a</code>          | Archive mode: preserves permissions, timestamps, symlinks, ownership |
| <code>-v</code>          | Verbose: shows each file being transferred                           |
| <code>-z</code>          | Compress data during network transfer                                |
| <code>--partial</code>   | Keep partially transferred files so rsync can resume after SSH drop  |
| <code>--progress</code>  | Show transfer progress for each file                                 |
| <code>--delete</code>    | Remove files in destination that no longer exist in source           |
| <code>--link-dest</code> | Hardlink unchanged files from previous snapshot (zero extra space)   |

|           |                                                          |
|-----------|----------------------------------------------------------|
| --exclude | Skip macOS system junk files (.DS_Store, .Trashes, etc.) |
|-----------|----------------------------------------------------------|

## 6. Restoring Backups

Each snapshot is a normal directory tree. Restore by running rsync in the other direction:

```
# Restore entire drive from a specific date
rsync -avz bunya:/QRISdata/Q1041/backups/snapshots_LaCie_Top/2026-03-26/ \
    /Volumes/LaCie_Top/

# Restore just one folder
rsync -avz bunya:/QRISdata/Q1041/backups/snapshots_LaCie_Top/2026-03-26/AHEAD/ \
    ~/Desktop/AHEAD_restore/

# Browse available snapshots
ssh bunya "ls /QRISdata/Q1041/backups/snapshots_LaCie_Top/"
```

## 7. Migrating Backups to Another Machine

Hardlinks are a filesystem-level feature. When copying the snapshot tree to a new location, you **must** use a tool that preserves hardlinks, otherwise every linked file becomes a separate copy and disk usage explodes.

### Tools that PRESERVE hardlinks:

- rsync -aHvz — the -H flag detects and recreates hardlinks at the destination
- tar — preserves hardlinks by default; stores data only once

### Tools that DO NOT preserve hardlinks:

- scp, cp (without flags), Finder drag-and-drop, Samba file copy

```
# Option 1: rsync with -H flag (can resume if interrupted)
rsync -aHvz bunya:/QRISdata/Q1041/backups/snapshots_LaCie_Top/ \
    /Volumes/NewDrive/snapshots_LaCie_Top/

# Option 2: tar on server, then download
ssh bunya "tar -cf /QRISdata/Q1041/snapshots_LaCie_Top.tar \
    -C /QRISdata/Q1041/backups snapshots_LaCie_Top"
rsync -avz --progress bunya:/QRISdata/Q1041/snapshots_LaCie_Top.tar \
    /Volumes/NewDrive/

# Verify hardlinks survived (inode numbers should match)
ls -li snapshots_LaCie_Top/2026-03-26/AHEAD/somefile.nii
ls -li snapshots_LaCie_Top/2026-04-15/AHEAD/somefile.nii
```

## 8. SSH Stability Tips

For long-running transfers, add these lines to your ~/.ssh/config under the bunya host block to prevent idle disconnections:

```
Host bunya
    HostName bunya.rcc.uq.edu.au
    User uqhsun8
    ForwardAgent yes
    ForwardX11 yes
```

```
ForwardX11Trusted yes
ServerAliveInterval 60
ServerAliveCountMax 3
```

Use `caffeinate -s` before the script command to prevent your Mac from sleeping during the transfer. If SSH does drop, simply re-run the same script — `--partial` ensures `rsync` resumes where it left off.

## 9. Verification Commands

```
# Check snapshot sizes (hardlinked snapshots should show small delta)
ssh bunya "du -sh /QRISdata/Q1041/backups/snapshots_LaCie_Top/*/"

# Check total actual disk usage (accounts for hardlinks)
ssh bunya "du -sh /QRISdata/Q1041/backups/snapshots_LaCie_Top/"

# List all snapshots
ssh bunya "ls -la /QRISdata/Q1041/backups/snapshots_LaCie_Top/"

# Verify the 'latest' symlink
ssh bunya "readlink /QRISdata/Q1041/backups/snapshots_LaCie_Top/latest"
```